Industrial Internship Report on

” IoT & Embedded Systems Architecture, Hardware Interfacing, and Scalable Deployment”

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by **upskill Campus** and **The IoT Academy** in collaboration with Industrial Partner **UniConverge Technologies Pvt Ltd (UCT)**. This internship focused on IoT and Embedded Systems core architecture, physical signal acquisition, hardware interfacing, and end-to-end industrial deployment strategies. The internship spanned four weeks of structured learning, practical exploration, and system design analysis. My project focused on understanding hardware-to-cloud IoT infrastructure, evaluating edge computing methods, sensor/actuator drive mechanisms, and designing scalable, secure M2M deployment architectures. This internship provided an excellent opportunity to gain direct exposure to industrial problems, hardware selection strategies, and scalable system design.

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1 Preface

This industrial internship report documents the technical knowledge, practical findings, and architectural models studied during the 4-week program in Internet of Things (IoT) and Embedded Systems. The report captures the transition from fundamental electronics and sensor telemetry to complex edge computing, message queuing, and cloud platform integration.

The 4-week industrial training program was structured progressively, starting with foundational IoT architectures and industrial use-cases in Week 1, shifting to hardware components, microcontrollers, and single-board computers like Raspberry Pi in Week 2, advancing into complex deployment requirements, database selection, and multi-layer security in Week 3, and concluding with fundamental analog/digital electronics, electrical measurements, and sensor/actuator interfacing in Week 4.

Participating in this program bridged the gap between theoretical computer science concepts and physical system execution. Gaining hands-on familiarity with IoT pipelines, edge computing strategies, and electronic measurement techniques significantly strengthened my practical engineering capabilities and shaped my interest in pursuing an industrial career in IoT systems.

I extend my sincere gratitude to the technical mentors and organizing teams at **upskill Campus** and **The IoT Academy**, alongside the industry experts at **UniConverge Technologies Pvt Ltd (UCT)**, who provided structured guidance, practical learning materials, and invaluable technical mentorship throughout this internship.

For juniors and peers stepping into the IoT and embedded domain, focus heavily on understanding the underlying hardware constraints and data pipeline architectures rather than just high-level applications. Bridging physical circuit design with reliable cloud communication is the key to building scalable, production-ready systems.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

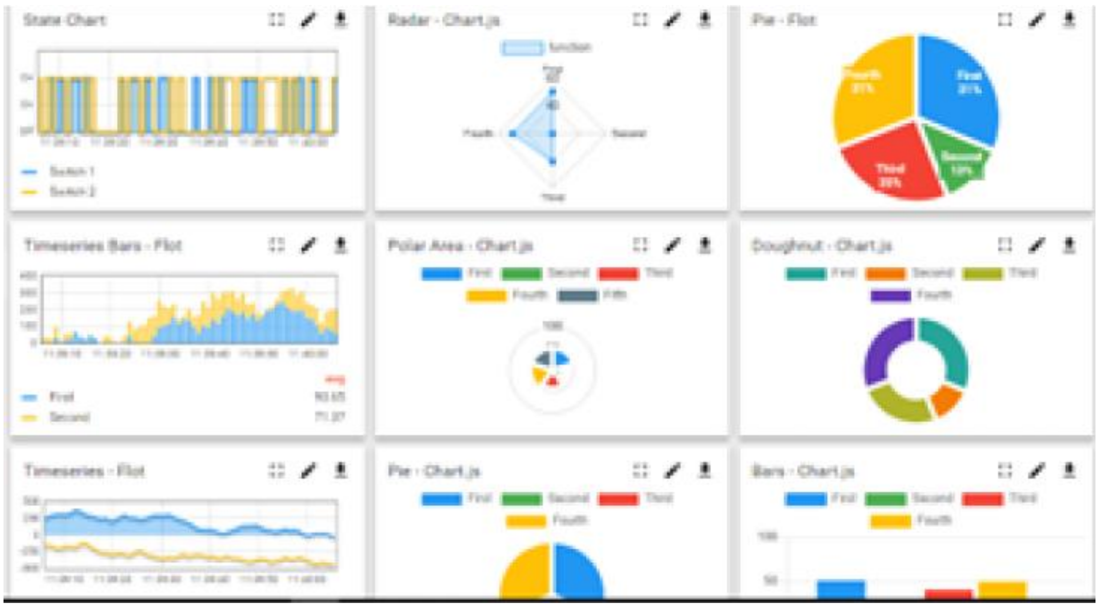
i. UCT IoT Platform ( )

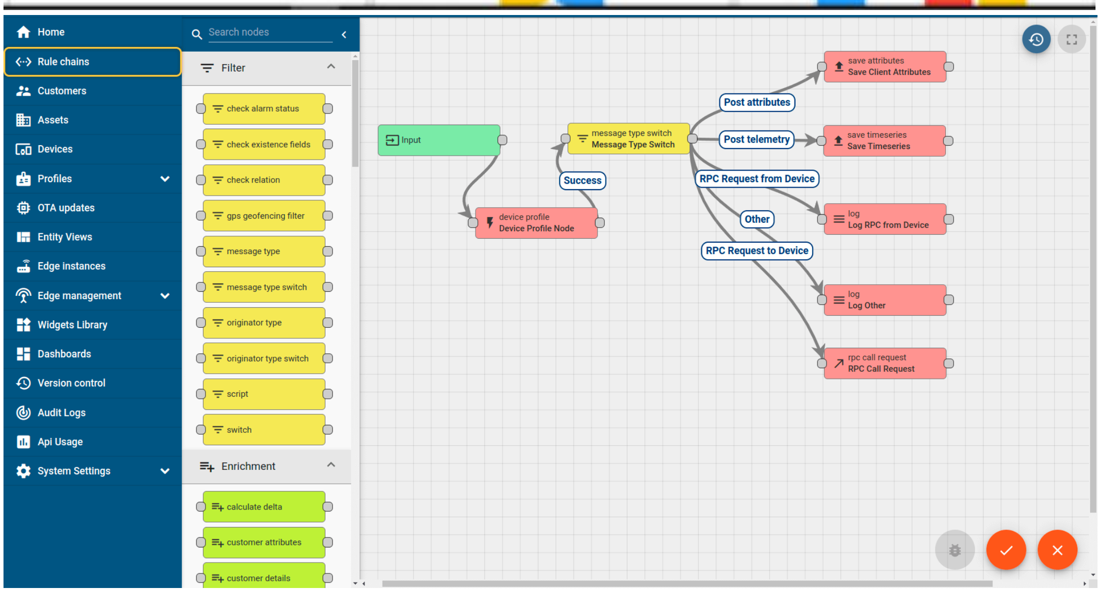
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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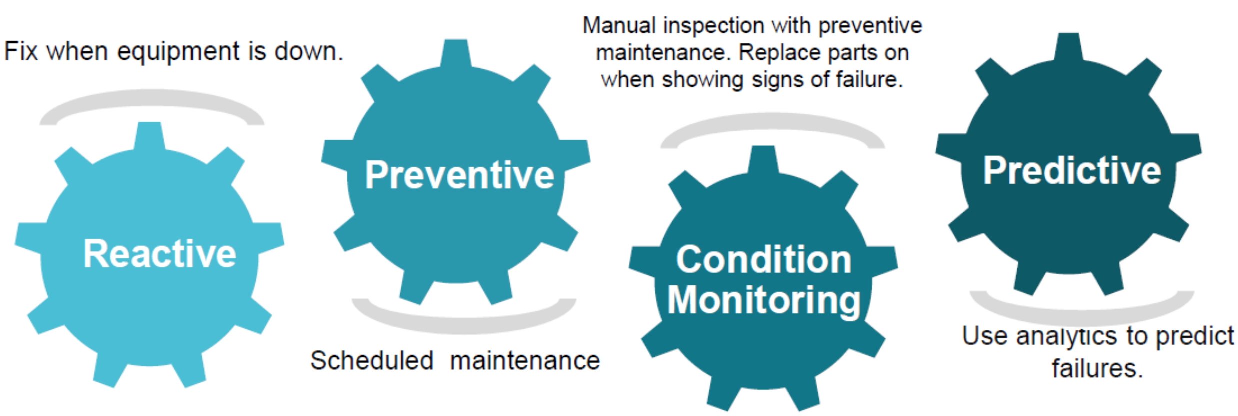
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

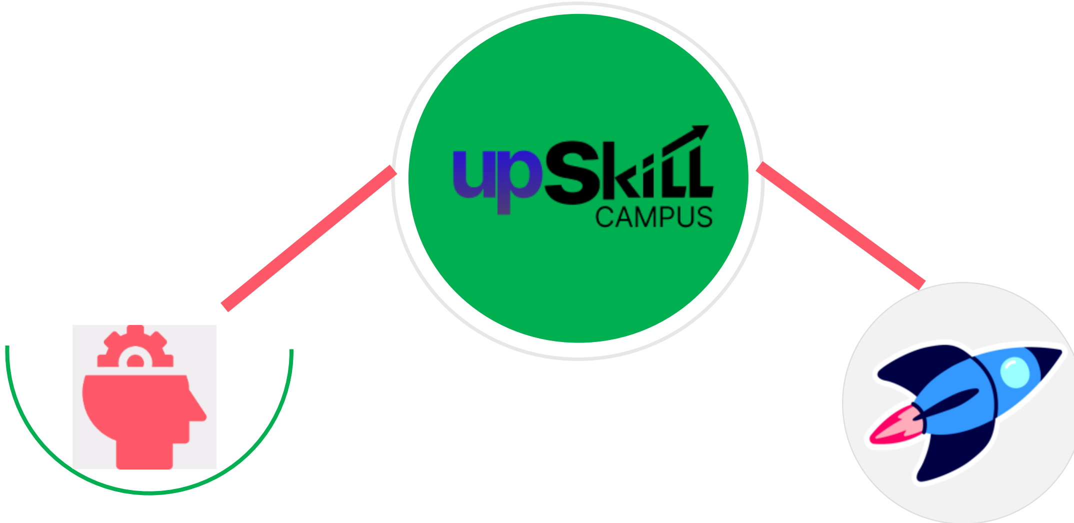
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/

2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1] upskill Campus & The IoT Academy Lecture Slides and Training Modules.

[2] Raspberry Pi GPIO & Hardware Specifications Documentation.

[3] Raspberry Pi GPIO & Hardware Specifications Documentation.

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
SBC	Single-Board Computer (e.g., Raspberry Pi)
GPIO	General Purpose Input/Output
LPWAN	Low-Power Wide-Area Network
M2M	Machine-to-Machine Communication
Kafka	Distributed Data Streaming & Queue Management Platform
OTA	Over-the-Air Software Updates

3 Problem Statement

Industrial IoT implementations face critical engineering hurdles that must be addressed during design:

1. **Data Volumetric Stress:** Streaming high-frequency raw telemetry continuously to the cloud creates massive network congestion and server overhead without local edge filtering.
2. **Hardware Selection Trade-Offs:** Choosing between low-power bare-market microcontrollers (AVR, ARM) and fully operating-system-driven single-board computers (Raspberry Pi) for edge telemetry nodes.
3. **Security Integration:** Safeguarding edge endpoints, network transmission channels, and central database storage against multi-surface threats without overburdening resource-constrained microcontrollers.

4 Existing and Proposed solution

FeatureTraditional / Existing ApproachProposed Industrial IoT SolutionData Flow Architecture

Raw, continuous telemetry streamed directly from endpoints to cloud databases.

Threshold-based local filtering and edge processing to aggregate telemetry before transmission.

Response Latency

High latency caused by continuous cloud round-trips for basic evaluations.

Low-latency local edge response complying with sub-100 ms sampling needs.

Ingestion Resilience

Direct database inserts leading to bottlenecks and dropped packets during traffic spikes.

Queue-managed streaming infrastructure utilizing Apache Kafka for high-throughput buffering.

Security Frameworks

Minimal perimeter-only security models with weak device authentication.

Multi-layered defense covering Device, Network, System, and User tiers.

4.1 Code submission (Github link):

https://github.com/abhiassistant/upskillcampus/blob/main/IoT_Sensor_Telemetry.cpp

4.2 Report submission (Github link) :

https://github.com/abhiassistant/upskillcampus/blob/main/IoT_Embedded_Abhishek_USC_UCT.pdf

5 Proposed Design/ Model

5.1 High Level Diagram (System Layers)

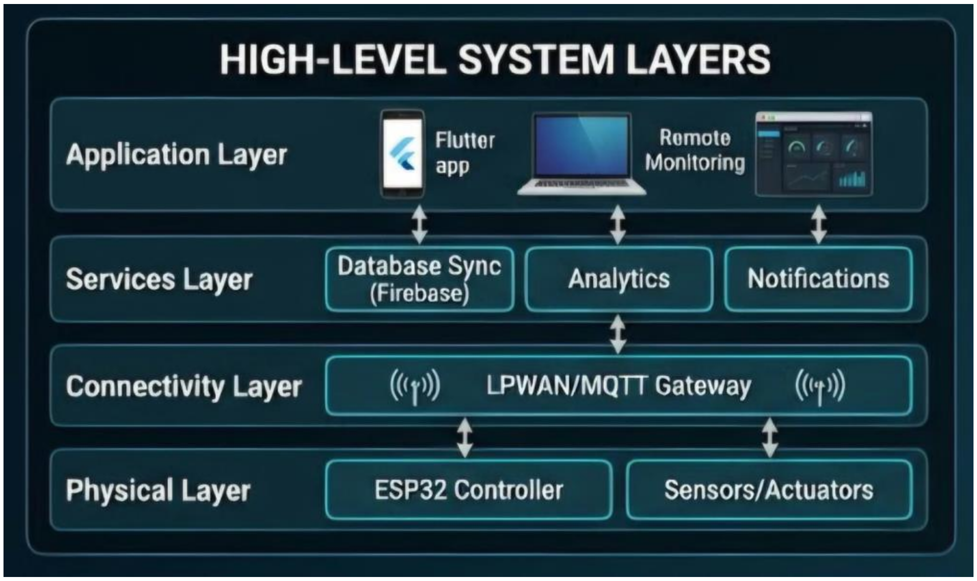


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (Hardware Interfacing)

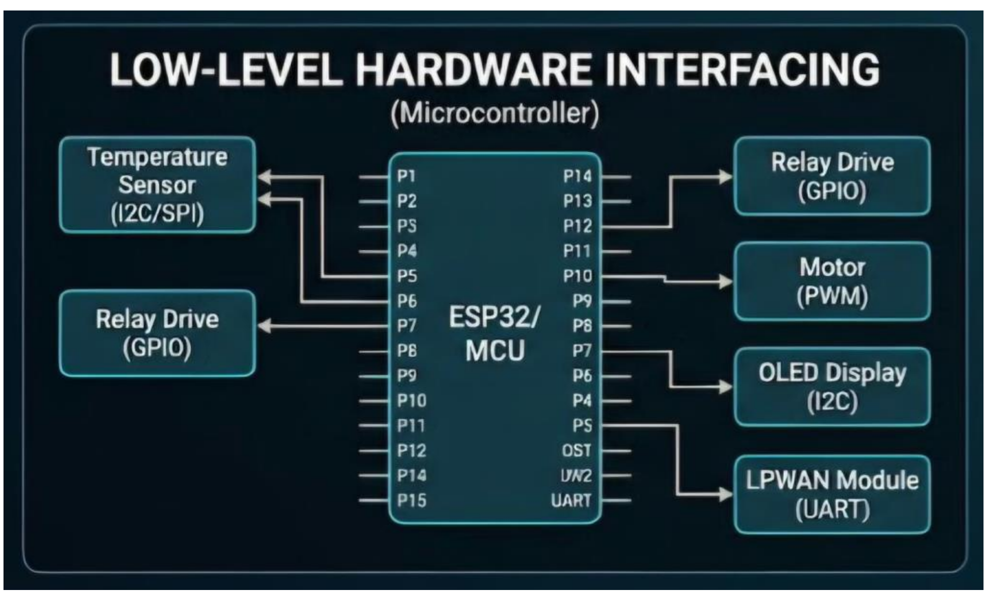


Figure 2: LOW LEVEL DIAGRAM OF THE SYSTEM

5.3 Interfaces

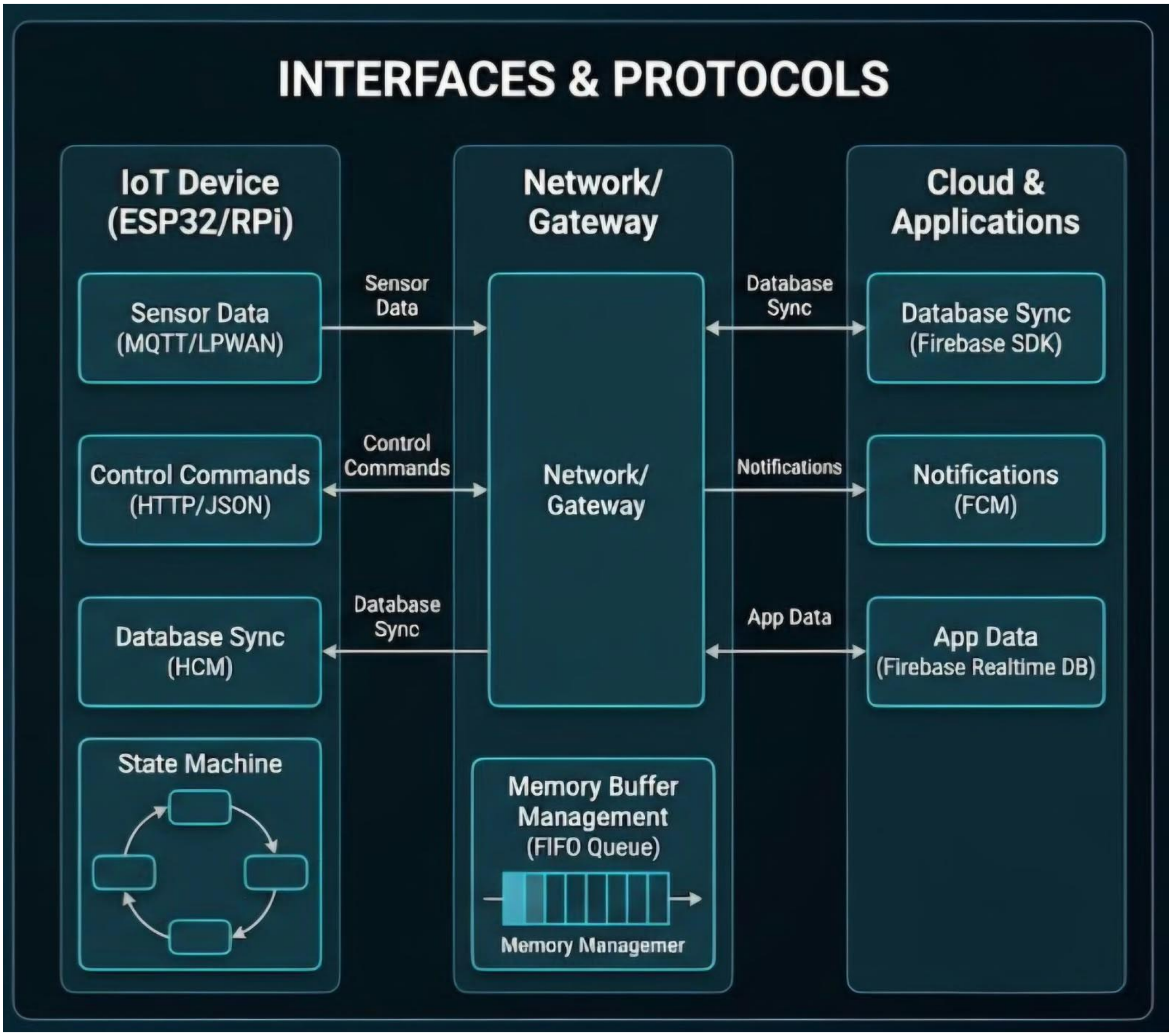


Figure 3: INTERFACE DIAGRAM OF THE SYSTEM

6 Performance Test

6.1 Test Plan/ Test Cases

- **Test Case 1:** Validate analog signal sampling accuracy and timestamp synchronization against sub-100 ms industrial performance thresholds.
- **Test Case 2:** Verify physical electrical measurement techniques, ensuring proper parallel voltmeter and series ammeter placement alongside unpowered resistance testing.
- **Test Case 3:** Test threshold-based communication configurations to measure payload reduction efficiency sent upstream to the server.

6.2 Test Procedure

- Conducted foundational electrical tests by applying Ohm's Law across passive and active components (resistors, capacitors, inductors, diodes).
- Configured test sensor telemetry routines (temperature, gas, distance) across microcontroller development targets.
- Evaluated gateway LPWAN range handling capability and message queue behavior under simulated high-volume data streams.

6.3 Performance Outcome

- Demonstrated optimized network bandwidth utilization by successfully replacing continuous telemetry streaming with threshold-triggered edge transmission.
- Established clear selection criteria separating low-power bare-metal microcontrollers for dedicated embedded sensing from Linux-based Single-Board Computers (Raspberry Pi) for intensive edge tasks.

7 My learnings

- **Embedded Electronics:** Mastered core physical measurement techniques (voltage across components in parallel, current through circuits in series, and unpowered resistance) alongside circuit stabilization methods like pull-up/pull-down resistors.
- **Hardware Architectures:** Acquired a comprehensive understanding of microcontrollers (AVR, ARM, PIC, MSP) and single-board computer capabilities (Raspberry Pi GPIO layouts, CPU architectures, memory configurations).
- **IoT Scalability:** Gained practical insights into processing high-volume telemetry through message queue management (Apache Kafka), time-series database selection, and edge computing data pipelines.
- **Security & Reliability:** Learned to design multi-layered security architectures protecting edge devices, LPWAN transport links, and cloud database repositories.

8 Future work scope

- **Physical Prototype Assembly:** Implement hands-on breadboard circuit builds integrating microcontrollers with active physical sensor arrays.
- **Edge AI Deployment:** Deploy lightweight machine learning models directly onto Raspberry Pi hardware to evaluate real-time local anomaly detection.
- **Optimized Distributed Storage:** Build and benchmark time-series database setups integrated with an active Apache Kafka message queue pipeline.